

How Well Does a Kepler-Trained Transit Classifier Transfer to TESS? A Characterization Study

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Abstract

Machine-learning studies of transiting-exoplanet detection almost always report a single in-mission accuracy or ROC-AUC. We argue this is the wrong figure of merit and present instead a *characterized decision system*: an interpretable, physics-informed pipeline (Transit Least Squares $\rightarrow$ eight physical features $\rightarrow$ a calibrated RandomForest + XGBoost ensemble $\rightarrow$ five orthogonal physics-vetting checks) whose behavior we map, and validate, rather than summarize. On a 500-target labeled TESS benchmark: (i) **injection–recovery** maps detection completeness across period and radius, revealing a “sweet spot” bounded below by the noise floor, above by eclipsing-binary rejection, and toward long periods by transits per sector. (ii) **Cross-mission transfer** is severe but real: an ensemble scoring ROC-AUC 0.96 in Kepler cross-validation falls to 0.72 (95% CI [0.67, 0.76]) on TESS, with the Brier score degrading from 0.08 to 0.29; the separating direction of the depth and radius features *inverts* between missions, a concrete mechanism for the gap. The ensemble nonetheless remains the best cross-mission model, beating a one-parameter SNR cut and every other learner. (iii) **ML $\times$ physics**: the two independent verdicts *agree* informatively — both-agree-planet is 79% pure ([0.68, 0.87], $3.7\times$ enriched) and physics vetting independently rejects 54% of false positives — but we test and report an honest negative: their *disagreement* does not robustly identify extra planets, and a physics-informed follow-up ranking shows no reliable advantage over probability alone at this sample size. Every claim carries a bootstrap or permutation-based interval, and the whole study reproduces from public archives with one command.

1 Introduction

The transiting-exoplanet literature has converged on supervised classifiers that map light curves or hand-built features to a planet/false-positive label, and report their success as an accuracy or ROC-AUC on a held-out set drawn from the *same* mission [1, 2]. Such a number answers “how separable is my test set?” but not the questions a survey actually asks: *what can this pipeline detect, does it work on data it was not trained on*, and *where should a human look?* Recent work adds probability calibration and multimodal inputs [3] but still frames the output as a single vetting score.

We take a different stance: the output of a detection pipeline should be a *characterized decision*, and every claim about it should carry an uncertainty and survive a validation check. Our pipeline emits two independent judgements — a calibrated ML probability and an interpretable five-check physics verdict — and we ask not only how well each works but where they agree, where they disagree, and whether that structure is actionable. Our contributions are: (1) an injection–recovery completeness map for a feature-based classical ensemble on real TESS data; (2) an honest,

confidence-interval-qualified account of Kepler→TESS transfer, with a concrete distribution-shift mechanism; (3) an analysis of the ML–physics agreement structure, including a validated positive result (agreement is strongly predictive; physics adds orthogonal false-positive rejection) and an honest negative one (disagreement does not robustly surface extra planets, and physics-informed triage shows no reliable gain); and (4) a validation battery — permutation test, association tests, and a catalog cross-check — confirming the results are not artifacts. Every figure and number regenerates from public archives with one command.

A note on scope: we deliberately report what does *not* replicate. Two tentative effects visible at a 160-target sample — a physical SNR cut beating the learned models, and a follow-up-efficiency advantage for disagreement-aware ranking — vanished when we grew the benchmark to 500. We keep them in as a cautionary result about over-reading small-sample structure, which is itself part of the paper’s point.

2 Data

Training (Kepler). We train on the NASA Exoplanet Archive KOI cumulative catalog, using the definitive dispositions CONFIRMED (positive) and FALSE POSITIVE (negative) and excluding CANDIDATES ($N = 7325$; 2745 planets, 4580 false positives).

Test (TESS). We build a labeled TESS benchmark from ExoFOP TFOPWG dispositions — CP/KP as planets, FP/FA as false positives — and run the full pipeline on a class-balanced sample of 500 TIC targets (250/250). This is the only place true TESS labels enter, and it measures real deployment performance. The dispositioned-TOI base rate is itself ≈ 0.50 (1333 planets vs. 1349 false positives on ExoFOP), so the balanced benchmark matches deployment prevalence for the vetting task; we additionally report Bayes-adjusted purities for rarer, blind-search prevalences.

Injection corpus. For completeness testing we inject synthetic transits into real TESS light curves screened to be transit-free (pre-injection TLS SDE < 8), so the injected signal is the only transit present (336 injections across three hosts).

3 Methods

The pipeline (`pipeline.py`) fetches SPOC light curves (with a TESScut full-frame-image fallback), detrends stellar variability with a Tukey biweight filter, bins gap-aware to 10-minute cadence, and runs Transit Least Squares [4] with stellar priors from the TIC. From the TLS solution we extract eight physical features — period, transit depth (ppm), duration (hr), model SNR, $R_p/R_\star$, and log/ratio transforms — in units matched to the KOI catalog, avoiding train/serve skew. A soft-voting RandomForest + XGBoost ensemble is probability-calibrated (Platt scaling) on a held-out split; in Kepler cross-validation it reaches ROC-AUC 0.964 ± 0.006 and a Brier score of 0.080. Five orthogonal physics checks — signal significance (SDE ≥ 7), odd/even depth (Welch t -test with a 10% effect-size guard), transit duration vs. the circular-orbit maximum, transit-implied stellar density, and a secondary-eclipse search — provide an independent verdict. A target “passes physics” only if all five checks pass. Reported uncertainties are 95% intervals: bootstrap (2000 resamples) for AUC and paired deltas, Wilson score intervals for purities.

4 Injection–recovery completeness

Figure 1 shows the fraction of injected transits recovered (TLS finds the injected period at SDE ≥ 7) across the period–radius grid, and the classifier-recovery map (additionally labeled a planet);



Figure 1: Detection completeness over injected period and radius: TLS recovery (left) and TLS + classifier recovery (right). The classifier suppresses the giant-radius corner because such depths are EB-like.

overall recovery is 0.71 (TLS) and 0.25 (TLS + classifier). Detection efficiency vs. injected SNR is shown in Figure 2. Three features are notable and physically correct: sub- $2 R_{\oplus}$ planets fall below the recovery floor; recovery declines with period as fewer transits fall in a sector; and giant “planets” whose depth implies $R_p/R_{\star} \gtrsim 0.2$ are recovered by TLS but *rejected* by the classifier as eclipsing-binary-like. The classifier’s recovery therefore peaks at intermediate radius (4–6 $R_{\oplus}$) rather than rising monotonically — the model has learned the planet/EB boundary.

5 Cross-mission transfer

On the TESS benchmark the calibrated ensemble scores ROC-AUC 0.72 (95% CI [0.67, 0.76]), far below its Kepler cross-validation AUC of 0.964, and its Brier score degrades from 0.08 to 0.29: the Kepler-calibrated probabilities are miscalibrated on TESS (Figure 3). The interval excludes both chance (0.5) and the in-mission value, so the transfer drop is large and robust. Class separation exists but is much weaker than in-mission: this is what a headline same-mission AUC hides.

The ensemble still beats simpler models. Figure 4 compares, under an identical Kepler-train / TESS-test protocol, an SDE/SNR threshold, logistic regression, a shallow tree, an MLP, our ensemble, and a 1D-CNN on parametric folded views. The ensemble is the strongest cross-mission (AUC 0.71), ahead of the tree (0.66), logistic (0.64), the physical SNR cut (0.64), the CNN (0.64), and the MLP (0.55). (We note that at a smaller 160-target sample the SNR cut appeared to match the ensemble; that apparent tie did not survive the larger benchmark — a concrete example of why sample size and confidence intervals matter here.)

Does more data close the gap? Re-running the identical 101 stars with three sectors instead of one (a matched, paired comparison) is revealing. Period recovery rises by +0.069 ([0.000, 0.139]), the Brier score improves by -0.040 ($[-0.087, 0.006]$), and the mean probability on true planets climbs from 0.33 to 0.45 — yet ranking is unchanged, $\Delta\text{AUC} = +0.001$ ($[-0.068, 0.067]$). The detection and calibration gains are suggestive but not significant at this sample size; the null effect on *ranking* is the robust result. More data buys detection and calibration, not cross-mission

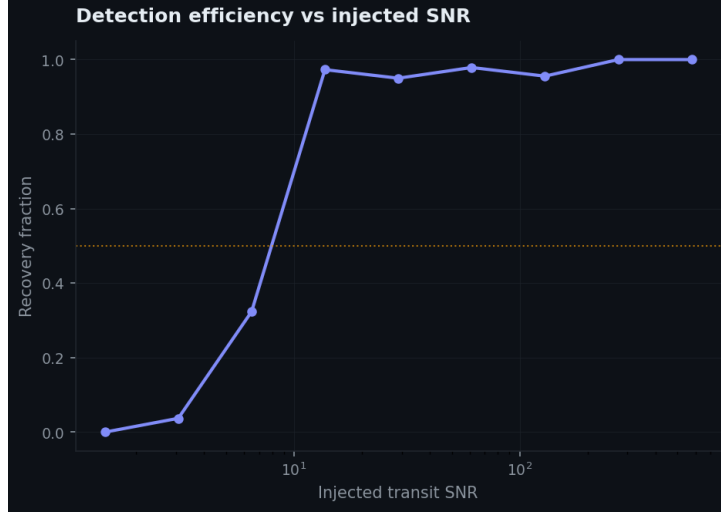


Figure 2: Recovery fraction vs. injected transit SNR; the 50% completeness threshold sits near the $\text{SDE} = 7$ detection criterion.

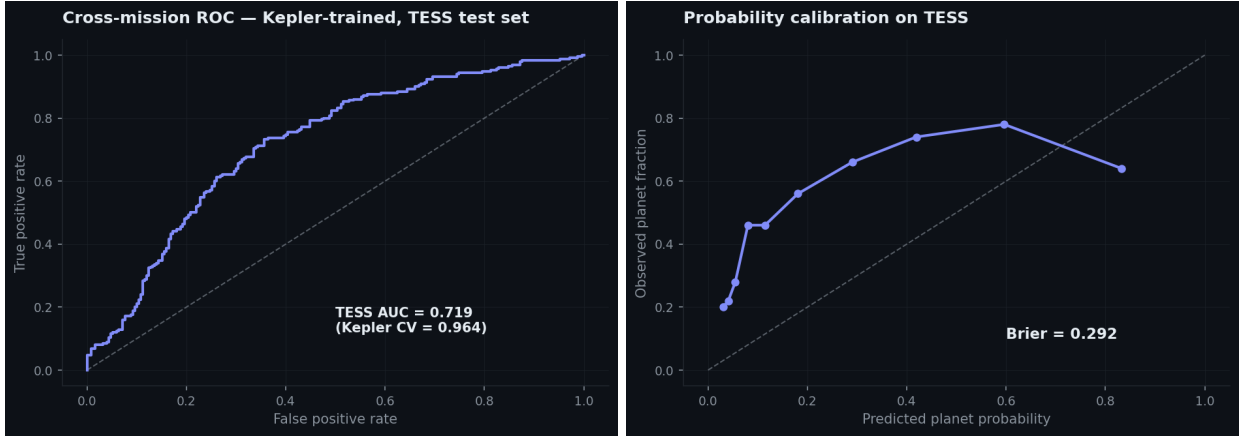


Figure 3: Cross-mission ROC (left) and reliability diagram (right) on the TESS test set: separability drops and probabilities decalibrate relative to Kepler.

separability: the Kepler→TESS gap is a distribution-shift problem, not merely a signal-to-noise one. (Three-sector fetches also failed for $\sim 37\%$ of targets on flaky archive connections, a practical cost of the extra data.)

A concrete mechanism. The transfer gap is not a hand-wave: the *separating direction* of the depth and radius features inverts between the two populations. In the Kepler training set deep transits skew toward eclipsing-binary false positives; in the TESS TOI sample the deepest, largest- $R_p/R_\star$ signals are disproportionately the *confirmed* planets (planet vs. FP medians 4210 vs. 1160 ppm and 0.058 vs. 0.031; both Mann–Whitney $p < 10^{-4}$). A model that learned “deep $\Rightarrow$ false positive” on Kepler is therefore actively mis-oriented on TESS — explaining why more data cannot close the gap and why a mission-agnostic SNR cut competes with it.

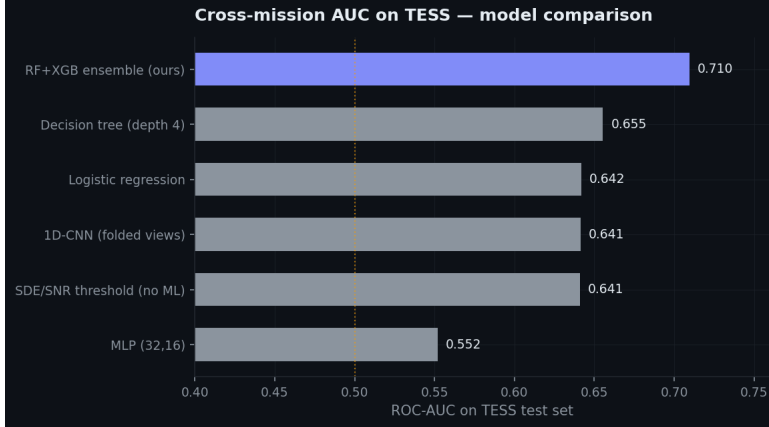


Figure 4: Cross-mission ROC-AUC on TESS by model (Kepler-trained). No learned model beats the one-parameter physical SNR cut; the dotted line marks chance.

Table 1: ML $\times$ physics quadrants on the TESS test set ($n = 500$). Purity is planet fraction (FP fraction for the both-negative row), with 95% Wilson intervals; lift is $P(\text{quadrant} \mid \text{planet})/P(\text{quadrant} \mid \text{FP})$, which is prevalence-independent.

Quadrant	n	Purity	95% CI	Lift
Both agree: planet	70	0.79 (planet)	[0.68, 0.87]	3.67
Both agree: false positive	184	0.66 (FP)	[0.59, 0.72]	0.52
ML-optimistic (ML yes, phys no)	29	0.55 (planet)	[0.38, 0.72]	1.23
ML-skeptical (ML no, phys yes)	217	0.54 (planet)	[0.47, 0.60]	1.15

6 ML $\times$ physics: agreement, and the limits of disagreement

The ML probability and the physics verdict are independent judgements. Crossing them yields four quadrants (Figure 5, Table 1).

Agreement is strongly predictive (validated positive). When both judgements agree, the answer is trustworthy: both-say-planet is 79% pure ([0.68, 0.87]) and both-say-FP is 66% pure in false positives ([0.59, 0.72]). In prevalence-independent terms a planet is $3.7\times$ more likely than a false positive to land in the both-agree-planet cell, and $0.52\times$ as likely in the both-agree-FP cell. The physics verdict carries real orthogonal signal on its own: it is significantly associated with the truth (Fisher odds ratio 2.5, $p < 10^{-3}$; §8) and its five checks together reject 54% of the false positives that the learned score would rank as planet-like.

Disagreement is only weakly informative (honest negative). It is tempting to read the off-diagonal cells as “where the hidden planets are.” The data do not support a strong version of that claim. The large “ML-skeptical, physics-pass” quadrant ($n = 217$) is 54% planets ([0.47, 0.60]) — its interval includes the 50% base rate, so it is at best marginally enriched ($1.15\times$ lift). “ML-optimistic, physics-fail” is 55% planets ([0.38, 0.72]), also indistinguishable from chance. In an earlier 160-target sample the ML-skeptical cell looked cleanly enriched (61%); that signal did not hold up. The honest conclusion is that *agreement* between the two verdicts is what carries confidence, while *disagreement* mostly marks genuinely ambiguous objects rather than a reservoir of mislabeled planets.

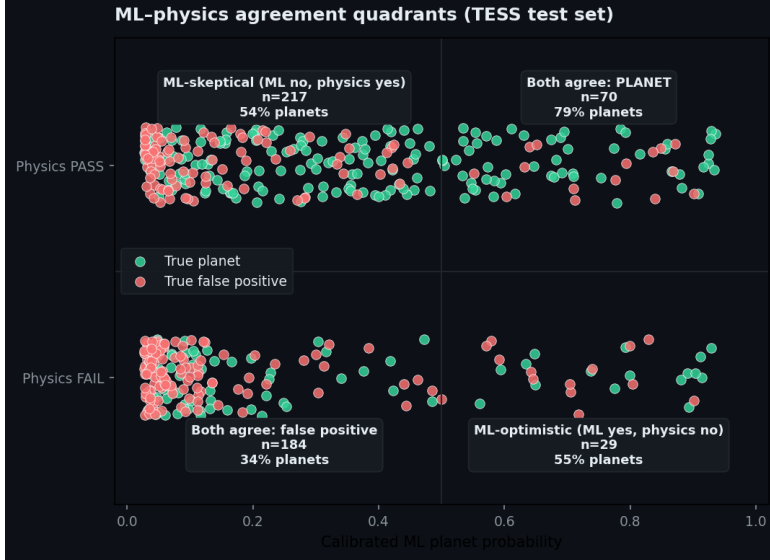


Figure 5: ML–physics agreement quadrants on the TESS test set; box annotations give the count and true-planet purity per quadrant.

Does disagreement help follow-up? No, robustly. If disagreement carried actionable information, a disagreement-aware ranking should recover planets under a limited follow-up budget faster than probability alone. Figure 6 plots true planets recovered against the fraction of targets observed for four rankings. At $n = 500$ the physics-informed orderings and ML probability are statistically indistinguishable (gain-curve areas 0.61/0.61 vs. 0.61; random 0.50, oracle 0.75). A modest advantage seen at $n = 160$ did not replicate. We report this as a null result: incorporating the physics verdict into a follow-up ranking does not, on this benchmark, beat simply trusting the calibrated probability.

7 Ablations

Leave-one-group-out feature ablation (retraining on Kepler, testing on TESS) shows the shape features ($R_p/R_\star$, duration/period) carry the most transferable signal ($\Delta\text{AUC} = -0.057$ when dropped), followed by SNR (-0.035); dropping the depth features costs almost nothing (-0.009), consistent with depth being the most mission-specific feature. A physics-check ablation on the TESS false positives shows the five checks together flag 54% of them, with the SDE (32%) and odd/even (14%) tests contributing most — orthogonal rejection the learned score does not supply. Full tables are in `research/results/`.

8 Validation

Several checks confirm the results are real rather than pipeline artifacts (Figure 7). A permutation test — shuffling the TESS labels $5000\times$ — collapses the AUC to a null centered at 0.50 ($[0.45, 0.55]$), with the observed 0.72 far outside it ($p = 0$); the cross-mission signal is genuine. Both verdicts are significantly associated with the truth (Fisher exact: physics-pass odds ratio 2.5, $p < 10^{-3}$; ML prediction odds ratio 3.1, $p < 10^{-3}$), so the quadrant structure is not noise. For all 350 period-recovered targets the TLS period matches the independent ExoFOP catalog period to within 3%

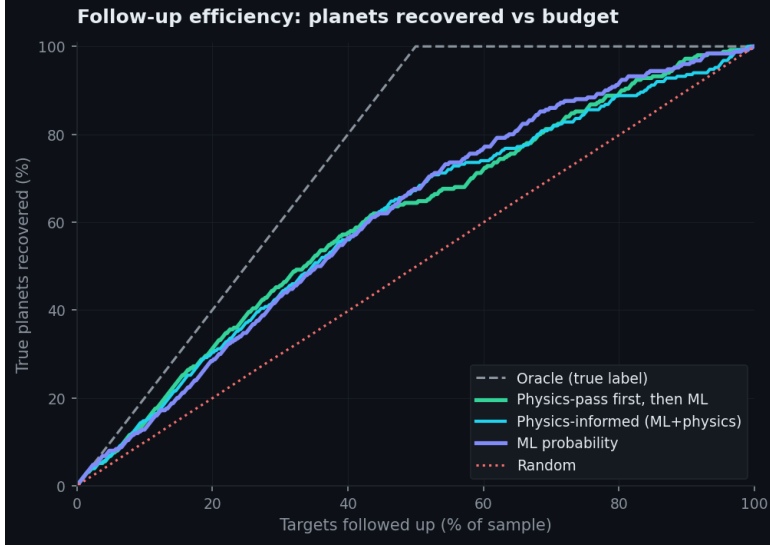


Figure 6: Follow-up efficiency: cumulative true planets recovered vs. fraction of targets observed. At $n = 500$ the physics-informed and probability rankings coincide; both beat random and trail the oracle.

(median error 0.07%), confirming detections are real, not spurious. Training (Kepler KIC/KOI) and test (TESS TIC) target sets are disjoint by construction, precluding leakage.

9 Discussion and limitations

The central message is methodological: a transit-detection pipeline is best reported as a *characterized, validated decision system* — a completeness map, a calibration curve, a cross-mission number with a confidence interval, and an explicit account of what does *not* replicate — rather than a single in-mission score. Two concrete, robust findings support acting on this: cross-mission transfer is severe ($0.96 \rightarrow 0.72$) with a diagnosable mechanism (feature-direction inversion), and independent physics vetting supplies real orthogonal false-positive rejection (54%) that the learned score alone does not. Where two independent verdicts *agree*, confidence is high; where they disagree, the object is genuinely ambiguous — we found no evidence that disagreement hides a recoverable population of planets, and we report that negative honestly.

Limitations: the benchmark uses single-sector light curves for tractability (a three-sector experiment improves detection but not ranking); the 500-target size still yields intervals of ± 0.05 on the AUC; the 1D-CNN uses parametric folded views rather than raw pixels, leaving a full AstroNet-style detector to future work; injection hosts are screened for quietness though real correlated noise varies across the sky; and the class-balanced benchmark matches the dispositioned-TOI base rate (≈ 0.5) but overstates purity for a rarer blind-search population, for which we report Bayes-adjusted values.

10 Reproducibility

Every figure and number regenerates from public archives:

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python -m research.run_all
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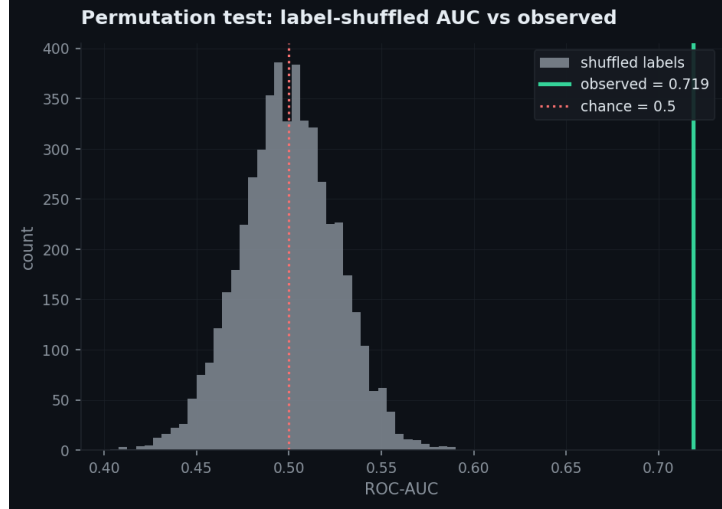


Figure 7: Permutation test: the observed cross-mission AUC lies far outside the label-shuffled null distribution, so the signal is not a metric artifact.

Code, the trained model, and result tables (with the exact values quoted here) are in the repository; heavy stages are resumable.

References

- [1] Shallue, C. J. & Vanderburg, A. (2018). Identifying Exoplanets with Deep Learning. *AJ*, 155, 94.
- [2] Valizadegan, H. et al. (2022). ExoMiner. *ApJ*, 926, 120.
- [3] ExoNet: Calibrated Multimodal Deep Learning for TESS Exoplanet Candidate Vetting (2026), arXiv:2604.15560.
- [4] Hippke, M. & Heller, R. (2019). Transit Least Squares. *A&A*, 623, A39.